

Sanjana K Rao

Senior Software Engineer

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Profile

Senior Backend Software Engineer with 6+ years of experience at large scale companies such as Target Retail Corp. and startups, building and scaling backend systems. Strong expertise in Java, Python, microservices, data quality systems, and distributed architectures, with a strong focus on system design, reliability, and performance. Interested in AI-driven systems and looking to work remotely.

Professional Experience

Target Corporation — Senior Engineer

Aug 2020 – July 2025

- Designed and operated high-throughput, event-driven distributed systems using Kafka and Springboot for processing millions of records daily with low-latency(< 100 ms), fault-tolerant data pipelines.
- Drove approximately \$4M in annual revenue impact by implementing Virtual Bundles to solve for inventory, sales and return issues. This enabled faster push of high-volume bundled items to market thus increasing conversion rates and repeat purchases.
- Improved barcode data integrity through checksum validation and anomaly detection using Python, Hive, and PySpark reducing revenue leakage by 10–15% across large-scale item data pipelines.
- Designed and implemented K-Means clustering techniques to automate item grouping on Target.com using Python and Scikit-learn reducing manual effort by 25%.
- Built high-throughput data quality pipelines using SQL and Internal tools to enforce consistency, accuracy, and timeliness across large-scale item data systems, supporting reliable downstream processing.
- Built Item Quality Index reporting and automated pipelines, improving data accuracy by 23% and reducing recurring data issues.
- Led design and implementation of a Spring Boot + Postgres system to automate item cluster rollout, replacing legacy Elasticsearch and eliminating key bottlenecks to improve system scalability and performance.
- Implemented production monitoring and alerting using Grafana, Kibana, and Prometheus improving system reliability and ensuring high-availability.

Tenovia — Data Engineer

2020

- Built data ingestion pipelines using Facebook Ads API to extract and load data into Snowflake cloud data warehouse.
- Worked with AWS services including S3, Lambda, and Athena for data storage, processing, and analytics.
- Assisted in data transformation and reporting workflows to support business analytics use cases.

Technical Skills

Languages: Java, Python, Kotlin, C++

Databases: PostgreSQL, MongoDB, Elasticsearch

Frameworks: Spring Boot, REST APIs, GraphQL

Data: Apache Kafka, Hive, PySpark, Pandas

Cloud: AWS- S3, Lambda, Athena, Snowflake

AI/ML: NLTK, TensorFlow, Claude code, Codex

Core Engineering: OOPs, Distributed Systems, CI/CD, Microservices, System Design, Event-driven architecture

Projects

Foodnote — iOS Application — Swift, SwiftUI, Core Data, Google Maps API

2026

- Built an iOS application using SwiftUI and Core Data to capture and manage structured dish data, enabling efficient local data persistence, querying, and location-based organization of food experiences.

Pokemon Game — Web Application — Python, Flask, Poke API, Supabase DB, Google Analytics

2026

- Developed an interactive Python application with API-driven gameplay and a persistent leaderboard using Supabase, handling real-time user interactions and analytics tracking.

Education

B.E. in Computer Science Engineering — PES University, Bengaluru - 8.5/10 CGPA

2016 – 2020